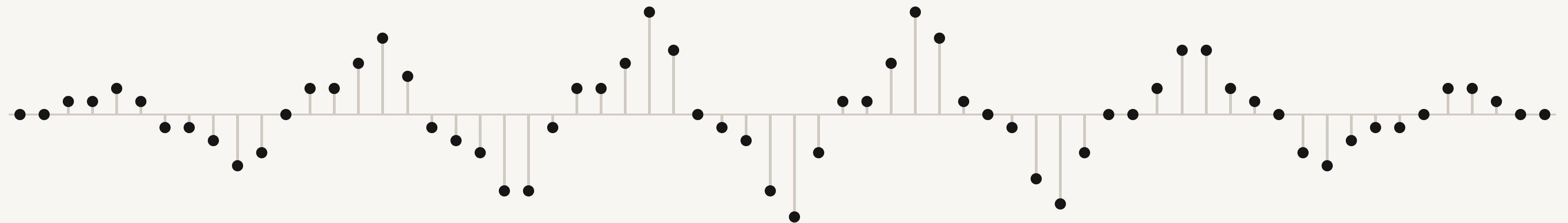


From ADPCM to AI

The Past, Present and Future of Game Audio Codecs

Collin Schupman, Senior Engineer

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About me

I've worked on game audio in commercial engines and proprietary ones.

Riot Games	Unreal Audio, Wwise Unannounced R&D
High Moon Studios	Proprietary engine tech Call of Duty: Modern Warfare 4, Black Ops 6 and Black Ops 7
Gas Powered Games	Unity and Wwise Unpublished title

Why this talk

I've found codec tech can be intimidating, especially on teams without a large audio programming group. Yet it has a real impact on the project.

This talk aims to build that understanding by going over:

How they work	Raw PCM audio, and what a codec does to shrink it
The problems	What kinds of problems codecs solve: memory, disk, voice chat, streaming and hardware limits
The history	How they came up, from ADPCM to Opus
Today	How they're used now: Opus, Vorbis, AAC and the platform codecs
The near future	How they might be used next: APAC, QOA and neural codecs

You leave with a better understanding of codecs, and a way to apply it to your own project.

A tradeoff in the wild: Titanfall

48 GB

PC install size, 2014

35 GB

of it was uncompressed audio

≈ 73%

of the install

The choice	Ship the PC audio uncompressed
The reason	Decoding costs CPU, and the minimum-spec PC had two cores
The trade	Disk space and download size, spent to keep CPU free for the game

“On a two-core [machine] with where our min spec is, we couldn't dedicate those resources to audio.”

Richard Baker, lead engineer at Respawn, to Eurogamer. Via The Escapist, March 2014.

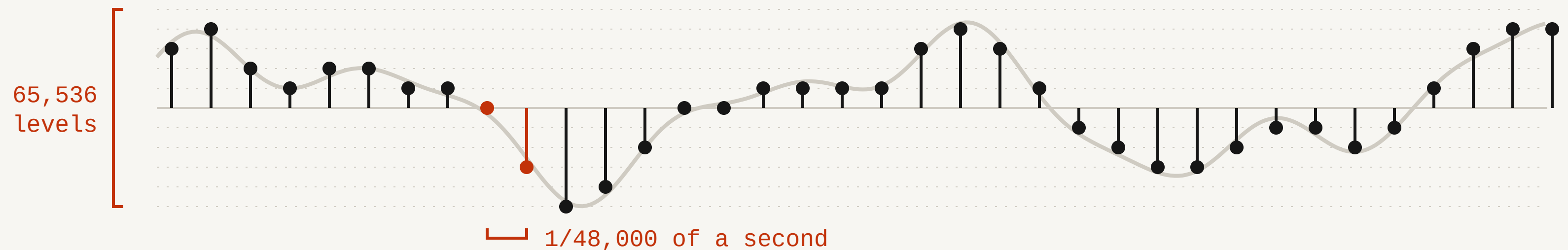
01

The basics

Raw PCM, how codecs work and the problems they solve

Raw PCM audio

PCM is sound stored as plain numbers: one measurement of the wave per sample, per channel. Nothing is modeled or thrown away.



Sample rate

How often the wave is measured:
48,000 times a second in most games.

Bit depth

How precisely each measurement is
stored. 16 bits gives 65,536 levels.

Channels

How many waveforms play at once.
Stereo is 2, and 5.1 is 6.

The cost of PCM: memory

48,000

×

16

×

2

=

1,536 kbps

samples per second

bits per sample

channels

uncompressed stereo, or 192 KB every second

Content	Size as PCM
One minute, stereo	11.5 MB
One hour, stereo	691 MB
One hour, 5.1	2.07 GB

For scale: all the sound RAM a PlayStation 2 game had



That RAM would hold about 11 seconds of this audio. Then multiply by every variation, language and platform.

The cost of PCM: streaming

Streaming means two things in games: reading audio from disk as it plays, and sending it over a network. Raw PCM makes both harder.

From disk

192 KB/s

per stereo stream, read constantly

- Competes for I/O with textures and level loads
- Hard drives and optical discs pay for every seek
- Each stream holds a RAM buffer: 2 s is 384 KB
- Nothing shrinks on disk: Titanfall shipped 35 GB

Over the network

768 kbps

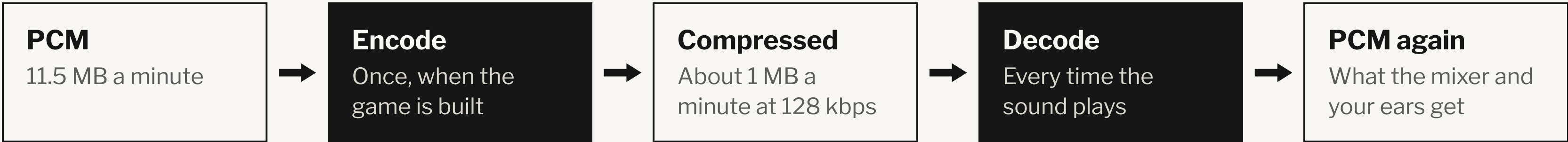
per mono voice, sent constantly

- Upstream bandwidth is shared with game traffic
- A 20 ms frame (1,920 bytes) overflows one packet
- No built-in recovery: a lost packet is a gap
- Fixed bitrate: it cannot adapt to a bad connection

From disk, PCM buys you zero decode CPU. Over the internet, that is rarely worth the bandwidth.

Codec basics

A codec is a coder and a decoder. It makes audio smaller to store or send, then turns it back into PCM to play.



Lossless	Every sample comes back exactly. Modest savings.	FLAC
Lossy: predictive	Predicts each sample and stores a rough correction. Cheap to decode.	ADPCM
Lossy: perceptual	Drops what the ear won't notice. Much smaller, but costs CPU to decode.	Vorbis, Opus, AAC
Bitrate	Data per second of sound, in kbps. Lower is smaller, until you start to hear it.	

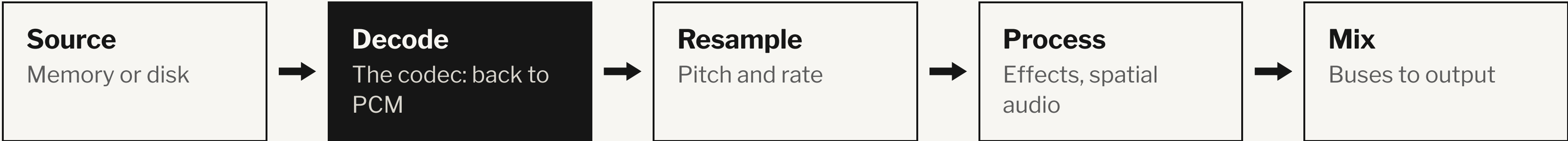
Every codec pays for its savings in CPU, latency or quality.

What a codec setting affects

Most projects pick a codec once, from a dropdown. That one setting decides:

File size	Install footprint and patch size
Streaming	Disk and I/O bandwidth under load
Memory and CPU	Resident RAM, and decode cost per voice
Platform compliance	Format requirements and certification
Sound quality	What players actually hear

The path of one voice



Where the source lives

Decoded in memory	PCM sits in RAM. No decode cost, most memory.
Compressed in memory	Decoded as it plays. Less RAM, CPU per voice.
Streamed from disk	Read in chunks, then decoded. Least RAM; adds I/O and latency.

Decode runs once per active voice, every buffer.

Inside a perceptual codec



Masking	A loud sound hides quieter ones near it in frequency and time. Those get fewer bits.
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Encode	At build time. Slow and thorough, and it runs once.
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Decode	At runtime. The same steps in reverse, per voice, every buffer.
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Sample-based audio

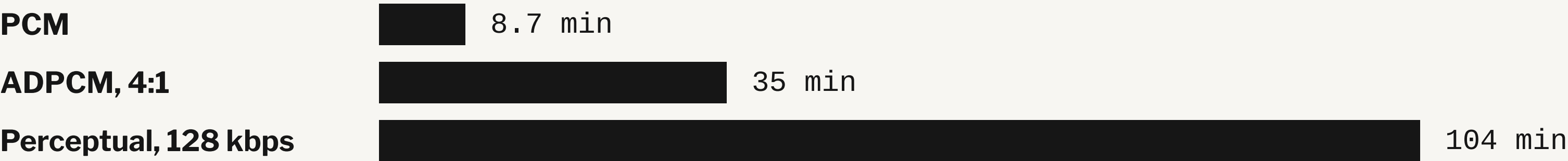
Early consoles synthesized sound on a chip. Once games played back recordings, the sample data had to fit in very little memory.

Year	Hardware	Audio memory	Compression
1990	Super Nintendo	64 KB audio RAM, 8 voices	BRR, an ADPCM variant, about 3.5:1
1994	PlayStation	512 KB sound RAM, 24 voices	4-bit ADPCM; XA-ADPCM streams from CD
1996	Nintendo 64	Samples share the cartridge ROM	ADPCM
2000	PlayStation 2	2 MB sound RAM, 48 voices	4-bit ADPCM
2005	Xbox 360	Shared system RAM	XMA, decoded in hardware

ADPCM won that era because the hardware could decode it for almost nothing.

Memory and disk

Minutes of 48 kHz stereo that fit in 100 MB



Dialogue	10 hours of mono VO is 3.5 GB as PCM. At 48 kbps Opus it is about 216 MB.
Languages, platforms	Each one multiplies the total, on disk and in every patch.

Voice chat

Live speech has to go both ways, in real time, over a connection that drops packets.

768 kbps → ≈ 24 kbps

mono PCM at 48 kHz Opus speech, 32× smaller

Low bitrate	Upstream bandwidth is shared with the game's own network traffic.
Low delay	Frames of 20 ms or less, so conversation stays natural.
Packet loss	The decoder conceals missing frames. Forward error correction fills gaps.
Lineage	Speex (2003), then SILK and CELT, merged into Opus (2012).

Codecs in streaming

In a game, streaming reads a file from disk at a bitrate you chose. Over a network, the codec has to adapt to the connection.

Where	Codecs	What the codec has to handle
Music services	Vorbis, AAC, Opus; FLAC or ALAC for lossless tiers	Quality per bit, and a decoder on every device
Video on demand	AAC family, including xHE-AAC; Dolby Digital Plus	Adaptive bitrate: several encodes, switched per segment
Live streams	AAC	Real-time encoding on the streamer's machine
Voice and calls	Opus	Low delay, and recovery from lost packets
Cloud gaming	Usually Opus	A delay budget shared with video and input

Three codecs carry most of it: Opus, Vorbis and AAC.

Hardware

Several codecs exist because of a specific hardware limit.

Dedicated decoders	Sound chips and DSPs decode ADPCM, XMA or AAC, so the CPU doesn't have to.
Battery	Hardware decode draws less power, which matters on handhelds and phones.
Storage media	Optical discs seek slowly, so music and VO stream as compressed data. Cartridge and download sizes cap the total.
Output links	Optical S/PDIF carries only stereo PCM, and Bluetooth is narrower still. Both need a codec on the way out.

02

The past

A short history of game audio codecs

A short history

1973

ADPCM

Predictive coding from Bell Labs. Still in every engine.

1993

MP3

Perceptual coding goes mainstream.

1997

AAC

MP3's successor, later decoded in phone hardware.

2002

Vorbis

Royalty-free. Becomes the game default.

2005

XMA

Xbox 360's format, decoded in hardware.

2011

ATRAC9

Sony's low-latency codec, from PS Vita on.

2012

Opus

IETF standard. Speech and music in one codec.

Predictive coding first, then perceptual coding, then platform-native codecs, then one flexible open codec.

ADPCM and MP3

Why it survives

ADPCM

- Fixed 4:1 against 16-bit PCM
- Near-zero decode CPU
- Sample-accurate seeks and loops
- Still right for short, busy SFX

Why it faded

MP3

- Encoder delay and padding add silence
- Seamless loops break
- Patent licensing until 2017
- Vorbis did the job for free

Still in use

PlayStation emulators still decode the console's ADPCM samples. A fun read on its sound chip:

jsgroth.dev/blog/posts/ps1-spu-part-1

03

The modern toolbox

Opus, Vorbis, AAC, XMA, ATRAC9, and how to choose

Codecs in use today

Codec	Good for	Watch out for	Wwise / Unreal
ADPCM	Short, frequent SFX	Fixed ~4:1; hiss on bright sounds	Both
Vorbis	Music, ambience	Weaker than Opus at low rates	Wwise
Opus	VO, music, low bitrates	More decode CPU than ADPCM	Both
Bink / RAD	Unreal's default	Unreal only	Unreal
AAC	Apple targets	Licensing; loop padding	Wwise (Apple)
XMA	Xbox, hardware decode	Xbox only	Wwise; UE Platform Specific
ATRAC9	PlayStation, cheap decode	PlayStation only	Wwise; UE Platform Specific

Support varies by version and platform. Check your build.

Vorbis and Opus

The incumbent

Vorbis

- Royalty-free; 1.0 released in 2002
- Long-time default in Wwise and many engines
- Mature encoders, predictable decode
- Falls behind at low bitrates

The challenger

Opus

- IETF standard, RFC 6716 (2012)
- SILK for speech, CELT for music
- 6 to 510 kbps; frames from 2.5 to 60 ms
- Best quality per bit at low rates

Default new streamed content to Opus. Keep Vorbis where a pipeline already depends on it.

Platform codecs

Codec	Platform	What you get	What you give up
XMA	Xbox	Hardware decode; CPU freed for more voices	Xbox only
ATRAC9	PlayStation	Low latency and low decode cost	PlayStation only
AAC	Apple, Android	Hardware decoders on most phones	Licensing; loop padding

Each platform gets its own encode. Budget build time and QA for it.

Codecs you didn't choose



Bluetooth headsets	SBC, AAC, aptX, LDAC, and LC3 on LE Audio. Adds latency too.
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TVs and soundbars	Bitstream output encodes the mix live as Dolby Digital or DTS.
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Capture, streams, cloud	The final mix is encoded again, usually as AAC or Opus.
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Lossy generations stack, so leave headroom in the in-game encode.

**Choose a codec per asset
class, not per project.**

A decision order

Ask in this order. Stop at the first yes.

1	Short, frequent, latency-critical?	ADPCM, or PCM if memory allows
2	Dialogue or voice chat?	Opus, at speech bitrates
3	Long and streamed from disk?	Opus or Vorbis
4	CPU-bound at peak voice count?	Platform codec: XMA, ATRAC9, AAC

Then listen on target hardware, at real mix levels, before locking bitrates.

What to measure

Decode CPU	Profile at worst-case voice count, not one sound in isolation.
Memory vs. streaming	Resident assets want cheap decode. Streams want ratio and seeking.
Loops and seeking	Check that sample-accurate loop points survive encoding.
Bitrate per class	Set per asset class, then listen on target speakers and headphones.
Platform rules	Check each platform's format requirements.
Tools and licensing	Confirm encoder support in the pipeline, and any license terms.

04

The frontier

Apple's APAC, QOA, neural audio codecs and Opus 1.6

Apple's APAC

- Apple Positional Audio Codec. In the OS since iOS 13, formally defined with iOS 18.
- Carries channel-based and scene-based (Ambisonics) audio.
- Delivery format for Apple Spatial Audio Format (ASAF), shown at WWDC25.
- Required for Apple Immersive Video. Reported up to 768 kbps.
- Streams over HLS, with loudness and DRC metadata embedded.

For game audio

- Useful for linear content: cinematics, trailers, immersive video.
- Apple platforms only, encoded with Apple's OS tools.
- A delivery codec. Your engine still renders the interactive mix.

QOA: the Quite OK Audio format

3.2

bits per sample

5:1

fixed, against 16-bit PCM

≈ 400

lines of C for the reference codec

- An open, MIT-licensed lossy codec by Dominic Szablewski, released in 2023.
- Works in the time domain like ADPCM: it predicts each sample and stores a 3-bit correction.
- By its author's account: better quality than ADPCM, and 3× faster to decode than Ogg Vorbis.
- The format spec fits on one page.

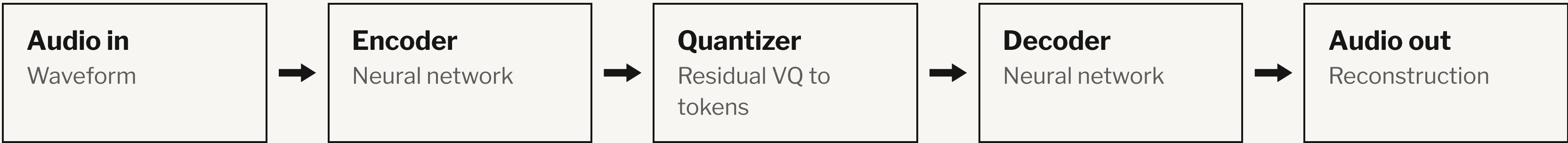
github.com/phoboslab/qoa

For game audio

- Ships in open-source engines so far: Godot (4.3 and later) and raylib.
- A candidate for the short, busy SFX where ADPCM is used today.
- For music and VO, Opus and Vorbis still get far smaller.

Neural codecs

A network learns the transform and the perceptual model from data.



Codec	From	Year	Notes
SoundStream	Google	2021	Streamable and causal; about 3 to 18 kbps
EnCodec	Meta	2022	General audio; about 1.5 to 24 kbps
Lyra	Google	2021–22	Speech only; v2 is built on SoundStream

Language models in codecs

Neural codecs turn audio into tokens, and a language model can predict tokens. That is used in two directions.

Approach	Example	What it does
LM as entropy model	EnCodec (2022)	A small Transformer predicts tokens; up to 40% fewer bits
LM fills in detail	LMCodec (2023)	Send coarse tokens; a Transformer generates the fine ones
Codec as LLM tokenizer	Mimi, in Moshi (2024)	12.5 frames/s, about 1.1 kbps; replies in about 200 ms
Ultra-low-rate speech	ZipCodec (2026)	0.8 kbps streaming speech from an 842M-parameter model
Promise	Speech near 1 kbps, and tokens an AI character can generate directly.	
Catch	The decoder generates plausible detail, so it can change a word or a voice. Most are trained on speech only.	

Barriers in games

Cost per voice	One model inference per stream, and a game mixes dozens of voices.
Latency	Many models need lookahead. Streaming variants trade quality for speed.
Training data	Mostly speech and music. Sharp SFX transients are unfamiliar.
Determinism	Float math on GPUs and NPUs can differ across platforms.
Model size	Weights ship with the game and compete for memory.
Tooling	Little middleware support today, and no standard bitstream.

Opus 1.6: neural tools inside a classic codec

9 kbps

fullband speech, matching Opus 1.4 at 18

600 kB

loss-recovery model, down from 1,800

96 kHz

Opus HD, experimental

- Released December 2025, and still fully compatible with the 2012 standard.
- A neural network rebuilds the top of the spectrum (8 to 20 kHz) from wideband speech.
- In the Opus team's own listening test, it beats a purely neural codec (EnCodec) at 9 kbps and above.
- Also new: a 24-bit API, and Opus HD for 96 kHz audio at up to 2 Mb/s.

opus-codec.org/demo/opus-1.6

For game audio

- Voice chat benefits first: better speech from the codec you already ship.
- The neural features are opt-in at build time, and cost extra decode CPU.
- Old streams and old decoders keep working.

Where neural audio shows up first

Now	ML inside classic codecs. Opus 1.5 and 1.6 added neural packet-loss recovery, speech enhancement and bandwidth extension.
Next	Low-bitrate speech. Research codecs reach 20 ms latency at roughly 20× real time on one CPU.
Speculative	Tokens in place of waveforms. Generative pipelines that store and stream codec tokens, decoded on device NPUs.

Expect it in voice before music or SFX.

Takeaways

- 1 Codec choice is a budget decision: CPU, memory, disk and quality at once.
 - 2 Decide per asset class. ADPCM for short SFX, Opus for VO and streams, platform codecs where hardware helps.
 - 3 On the frontier, APAC is for spatial delivery, QOA is a modern take on ADPCM, and neural codecs will reach voice first.
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Thank you

Questions?

Collin Schupman, Audio Programmer

`collinschupman@gmail.com`



LinkedIn



Email



GitHub